

ENHANCED OBJECT DETECTION ON AERIAL CITYSCAPES VIA AUGMENTATION AND YOLO VARIANTS

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ABSTRACT

This paper details our submission to the IEEE ICIP 2025 Grand Challenge on the Cityscape Aerial Image Dataset. This dataset includes top-down remote sensing images featuring a diverse range of objects, from large structures like buildings and basketball courts to smaller elements such as vehicles and crosswalks. A key characteristic of this dataset is a significant class imbalance. To address this, we propose an object detection framework that leverages advanced data augmentation techniques and state-of-the-art YOLO-based models (specifically, YOLOv11 and YOLOv12). Our core contribution is an augmentation strategy that balances class distributions and enhances object diversity. This is achieved by selectively removing, augmenting, and reinserting instances of minority classes, such as tennis courts and crosswalks. We present our model architecture, detail the training convergence, and provide a comprehensive error analysis. This targeted augmentation strategy proved highly effective, securing a top-2 ranking in the IEEE ICIP competition.

Index Terms— Aerial Object Detection, YOLO, Data Augmentation, Cityscape.

1. INTRODUCTION

Detecting objects in aerial images of urban landscapes is a challenge due to the highly diverse dataset, which includes objects of varying sizes, ranging from very small to very large. Moreover, the severe class imbalance makes object localization and recognition difficult. This work, prepared for the ICIP 2025 Challenge, addresses these issues through a robust data augmentation pipeline and variants of YOLO (YOLOv11 [1], YOLOv12 [2]).

In the field of object detection in aerial images, the DOTA dataset (Xia et al., 2018 [3]) is a large dataset of more than 2,800 high-resolution images (about 4,000 x 4,000 px), with 188,282 objects and VisDrone (Mao et al., 2020 [4]) has formulated the key to identify objects and their sizes. With their high scale and labeling quality, DOTA and VisDrone have become benchmarks for evaluating the performance of many deep learning models.

Methods such as RoI Transformer (Ding et al., 2018 [5]) and ReDet (Han et al., 2021 [6]) address orientation variation using an equivariant network. In contrast, TPH-YOLO (Zhu et al., 2021 [7]) added a Transformer-based prediction head to handle size variation due to different drone altitudes, along with a convolutional block attention model (CBAM) to enhance focus on high object density regions, achieving an AP of 39.18 % on VisDrone2021. However, minimal object detection is still weak due to class imbalance. To overcome this, AMRNet (Wei et al., 2020 [8]) used mask resampling to resample objects to balance the number of classes, and PENet (Tang et al., 2020 [9]) used Mask Resampling Module (MRM) to upsample classes with little data, combined two coarse and fine anchor-free detectors (CPEN, FPEN) with Non-maximum Merge (NMM) algorithm to improve small object detection on drone images, the best model achieved an improvement of 8.7% on VisDrone.

In terms of data augmentation, general techniques such as Mosaic (Bochkovskiy et al., 2020 [10]), MixUp, CutMix have proven effective on regular images. However, in drone images, ColMix (Ly et al., 2023 [11]) utilizes “image patching” to insert multiple objects into the background without masks, thereby enhancing object intensity more effectively than Mosaic and improving precision and recall. Nevertheless, it is sensitive to out-of-distribution noise. Messmer et al. (2021) [12] perform automatic image scaling preprocessing, which reduces the influence of measurement variation and speeds up inference by 2–3 times, with successful experimental results on UAVDT, VisDrone, and autonomous data. These methods are still random, which can introduce background noise and reduce realism.

Our core contribution is a novel, two-stage data augmentation process specifically designed to significantly enhance object detection in aerial cityscapes.

1. **Background Generation:** We meticulously remove small objects (e.g., vehicles, crosswalk markings) to create clean, rebalanced backgrounds that address inherent class imbalances.
2. **Object Augmentation and Reinsertion:** We then apply diverse transformations (rotation, cropping, color jitter) to these removed objects before intelligently pasting them back into the prepared backgrounds.

This innovative methodology generates richer, more varied training data. Our evaluations using Yolo variants on both orig-

This work was supported by the Ministry of Science and Technology’s project titled “Research and application of artificial intelligence technique to build the forecasting system of tropical cyclone activity over Bien Dong Sea and affecting Vietnam with lead time up to 3 days” (code: KC-4.0-46/19-25)

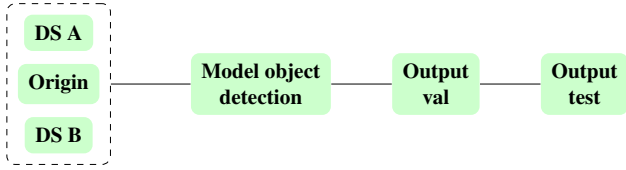


Fig. 1: Flow diagram of the object detection pipeline using DS-A, DS-B, and DS-Orig.

inal and augmented datasets unequivocally demonstrate the superior performance of our data augmentation workflows, leading to marked improvements in object detection accuracy on aerial cityscapes.

The paper is organized as follows: Section 2 presents the dataset, the method for object detection, and the main contribution in Section 3. Section 4 presents a deep analysis, detailed experience, and discussion. The last section, Section 5, is the conclusion.

2. DATASET

The primary data set used in this study is CADOT¹, which includes aerial cityscape images. This diverse data set is organized in the standard COCO format. It includes 3,234 images in the training set, 929 in the validation set, and 465 in the test set. The training set comprises 75,198 objects distributed across 14 different classes. The data set features objects of various sizes with a significant class imbalance. Table 1 (DS-Orig column) illustrates the distribution of these classes. Looking at the table, it is clear that the two classes, building and Small vehicle, are the two dominant classes with $\approx 40.9\%$ and $\approx 46.7\%$ of the total number of objects, respectively. The classes that can be considered average are medium vehicle ($\approx 5.1\%$), crosswalk ($\approx 5.0\%$), and large vehicle ($\approx 1.1\%$), and the rest account for less than 0.5% each.

Our augmentation aims to improve model robustness and generalization. Additionally, we minimize the amount of balancing so that the modeling is not too focused on high-volume objects.

3. TWO-STAGE AUGMENTATION METHODOLOGY

Our methodological approach is designed to systematically evaluate the impact of our proposed data augmentation strategies and the performance of different YOLO variants on the aerial cityscape object detection task. The core workflow, depicted in Figure 1, encompasses dataset preparation through augmentation, training of object detection models, and a comprehensive evaluation leading to the selection of the optimal configuration for the final test set.

3.1. Augmentation Strategy

In this section, we employ a specialized two-phase augmentation strategy. This strategy is crucial for addressing key challenges in

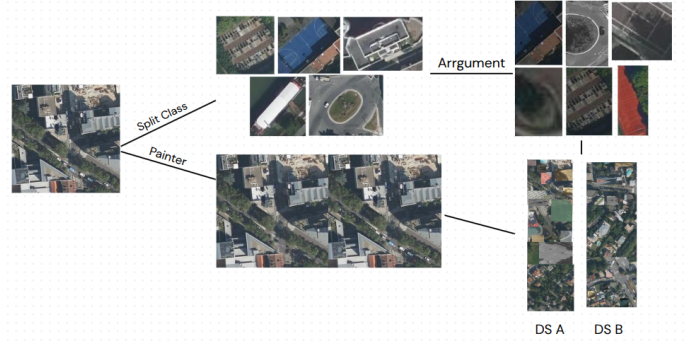


Fig. 2: Data augmentation for constructing DS-A and DS-B.

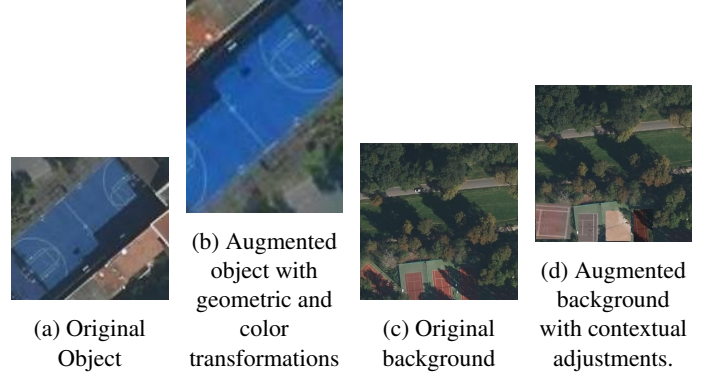


Fig. 3: Augmentation illustration

aerial imagery, such as significant class imbalance (e.g., numerous vehicles vs. fewer specialized structures) and the need to enhance the diversity of object instances, particularly for smaller or less frequent classes like "crosswalk," "ship," or "train." Specifically, We present two important strategies for data augmentation:

1. **Background Image Generation:** We first create enhanced background images. This involves selectively removing small, dynamic objects (e.g., 'vehicle,' 'crosswalk') from original images by using algorithms in [13]. The remaining static objects within these backgrounds are balanced to ensure a more uniform representation, preventing the model from becoming biased toward overly dominant background features.
2. **Object Augmentation and Re-insertion:** Concurrently, the removed objects and additional instances of critical classes undergo a series of augmentations. These include geometric transformations (rotations, cropping to 2/3 size to simulate scale variation) and photometric transformations (color jittering, saturation adjustments). These augmented objects are then carefully re-inserted onto the prepared background or suitable original images, creating rich and diverse training samples.

As illustrated in Figure 3, the proposed augmentation process generates data with substantial variation in object scale, perspective, and illumination. This enhancement directly addresses class imbalance by increasing the representation of under-sampled minority classes.

¹<https://cadot.onrender.com/dataset>

3.2. Experimental Datasets

The augmentation procedures for both datasets are illustrated in Fig. 2. Two augmented datasets, Dataset-A (DS-A) and Dataset-B (DS-B), are created from the original dataset (DS-Orig). DS-A is constructed by removing the majority-class objects and repeating the process three times. The resulting images are combined with two copies of the original dataset, and then minority-class objects are used to balance the representation. DS-B is created by replacing the easily recognizable objects in DS-Orig with more challenging and similarly sized objects, aiming to enhance the model’s ability to detect difficult-to-recognize classes. Specifically, The augmentation for DS-A prioritizes data balancing by interchanging objects within semantically coherent groups—such as “building,” “basketball field,” “football field,” “graveyard,” “playground,” “roundabout,” “swimming pool,” “tennis court,” “large vehicle,” “medium vehicle,” and “small vehicle”—to preserve the scene’s contextual integrity. Additionally, DS-B is constructed based on the analysis of errors identified during the initial training of the model. Its purpose is to strengthen the object classes that the model still recognizes well.

Due to their strict semantic and contextual rigidity, we did not augment the “Train” and “Ship” classes. Unlike the interchangeable objects in DS-A, these classes are context-bound (e.g., ships require water), and their inclusion would corrupt the semantic integrity of the augmented datasets.

Class	DS-Orig	DS-A	DS-B
Basketball field	15	22065	30
Football field	77	22437	154
Graveyard	130	22755	7951
Playground	44	22239	7779
Roundabout	37	22197	74
Swimming pool	60	22335	7811
Tennis court	98	22558	196
Ship	300	1800	600
Train	67	402	134
Large vehicle	841	37673	13399
Medium Vehicle	3865	46743	19446
Crosswalk	3751	11253	7502
Building	30764	30764	38455
Small vehicle	35149	35149	46865

Table 1: Class distributions.

The class distribution of both datasets is described in Table 1. DS-A exhibits a relatively balanced class distribution. In contrast, DS-B emphasizes rare and hard-to-detect objects, allowing for a focused assessment of model performance under challenging data scenarios.

4. EVALUATION AND DISCUSSION

4.1. Training and Evaluation Protocol

Each selected YOLO model (YOLOv11 and YOLOv12) was trained independently on three datasets: the original dataset (DS-Orig), the initially augmented dataset (DS-A), and the further refined augmented dataset (DS-B). Standard training procedures

were followed, including initialization with pre-trained weights (where applicable), the use of appropriate loss functions, and the application of a consistent learning rate schedule. The learning rate started at 0.01, with a linear warm-up phase during the initial epochs and a linear decay throughout the remainder of the training process.

We systematically assessed each configuration on an independent validation set using per-class Average Precision (AP) and mean Average Precision (mAP) at IoU thresholds of 0.5 and the [0.5:0.95] range. The final models² for the test set submission (Section 4.4) were selected based on a comprehensive analysis of validation accuracy, convergence behavior, and error patterns revealed in the confusion matrix.

4.2. Training Convergence

4.2.1. Convergence Discussion

The training plots illustrate the convergence behavior of different model configurations (YOLOv11, YOLOv12) across various datasets (DS-Orig, DS-A, DS-B).

Overall, all models demonstrate successful learning, characterized by decreasing loss values (box, class, and DFL) and increasing mAP scores over epochs. The learning rate schedules (described in Section 4.1) were consistent across all experiments, employing a cosine annealing strategy with a warm-up period to ensure comparable training dynamics.

A key observation is the performance difference between YOLOv11 and YOLOv12. Across all three loss metrics, YOLOv12 variants consistently achieved lower final loss values compared to their YOLOv11 counterparts on the same datasets. This superiority in minimizing loss generally translated to improved detection accuracy.

The impact of data augmentation (DS-A, DS-B) varied by model.

- For YOLOv11, the augmented datasets (DS-A, DS-B) did not lead to improved validation mAP scores compared to the original dataset (DS-Orig). In some instances, DS-Orig even showed slightly better or comparable loss and mAP metrics for YOLOv11 (e.g., Figures 4).
- Conversely, YOLOv12 appeared to benefit significantly from data augmentation. YOLOv12 trained on DS-A consistently yielded the lowest loss values across all types and achieved the highest mAP@50 scores. YOLOv12 on DS-B also outperformed YOLOv12 on DS-Orig in terms of mAP. This suggests that the YOLOv12 architecture is better able to leverage the diversity introduced by augmentation for improved generalization.

Validation losses generally tracked training losses, with small gaps indicating reasonable generalization. Some validation mAP curves show signs of plateauing towards the later epochs, suggesting that the models were approaching their performance limits under the current training regimen. The consistent learning rate schedules are applied to provide a stable baseline for these comparisons.

²Source code: https://github.com/hieupm123/HUS_ChapHet_CADOT_Challenge

4.2.2. Results and analysis

Figure 4 illustrates the progression of mAP@50 and mAP@50-95 over 100 epochs across three dataset configurations (DS-Orig, DS-A, and DS-B) with two model variants (YOLOv11 and YOLOv12). YOLOv12 on DS-B achieves a peak performance of mAP@50 $\approx$ 0.61 and mAP@50-95 $\approx$ 0.45, outperforming all other configurations. Transitioning from DS-Orig to DS-A (background enhancement + class balancing) improves YOLOv11 by approximately +0.15 mAP@50, highlighting the direct impact of balanced data distribution. DS-B, which emphasizes rare class re-synthesis and small object injection, further contributes with noticeable gains: +0.05 mAP@50 for YOLOv11 and +0.04 mAP@50-95 for YOLOv12 compared to DS-A.

Figure 4 also presents the Distribution Focal Loss (DFL), Class Loss, and Box Loss (train vs. validation), respectively. A clear narrowing of the train-validation gap is observed on DS-A and DS-B, whereas DS-Orig shows signs of overfitting after approximately 50 epochs (training loss continues to drop while validation loss plateaus). DS-A yields the lowest class and box losses, confirming the stabilizing effect of background normalization. Although DS-B exhibits higher absolute losses than DS-A, it ultimately achieves superior mAP scores, suggesting that the model tolerates greater localization errors in exchange for enhanced discrimination of hard-to-detect objects.

Overall, YOLOv12 + DS-B significantly improves detection performance (approximately +0.20 mAP compared to the baseline YOLOv11 + DS-Orig) without exacerbating overfitting. These findings highlight the crucial role of targeted data augmentation strategies in conjunction with model updates, providing a practical approach for enhancing detection performance on rare objects in urban remote sensing imagery.

4.3. Error Analysis

4.3.1. Error Discussion

- **Praiseworthy Highlights:** The model demonstrates considerable strength in identifying familiar classes with abundant samples. One cannot help but be impressed by the 8703 correct identifications for "building" and a whopping 9596 for "small vehicle". Even "crosswalk", with nearly 1000 correct predictions, is a noteworthy achievement. This suggests the model has "aced the test" for these common objects.
- **The "Field" of Woes:** It is pretty disheartening to observe the performance of field-related classes. "Basketball field" scored a solitary correct prediction – a figure that's frankly startling. "Football field" (8 correct) and "playground" (4 correct) do not fare much better. Distinguishing these open areas is a genuinely tough nut for the model to crack.
- **The Perennial Vehicle Size Saga:** Differentiating between "small vehicle", "medium vehicle", and "large vehicle" remains a significant, and somewhat predictable, headache. While "small vehicle" is a star performer, the confusion between "medium vehicle" and "small vehicle" (235 instances) and "large vehicle" (30 instances) – and vice versa – indicates that the model is still somewhat

muddled when estimating size. The 214 instances where "small vehicle" was mislabeled as "medium vehicle" are also not insignificant.

- **When 'Building' Becomes a Magnet:** Interestingly, the "building" class seems to exert a peculiar 'gravitational pull.' Several instances of "swimming pool" (7), and even "football field" (1) and "tennis court" (1), were mistakenly 'zoned' as "building" by the model. Is it possible that the model tends to 'take refuge' in this highly prevalent class when uncertain?
- **Commendable Efforts, Yet No Major Breakthroughs:** Classes like "graveyard" (49), "roundabout" (16), "ship" (41), or "train" (16) show the model is certainly trying, but the results remain modest. These are likely 'tough cases' demanding more finesse and distinct features for the model to latch onto.

In summary, this confusion matrix acts as a clear mirror, reflecting what the model has mastered and where it still stumbles. While there are proud highlights, especially with common objects, the journey to conquer challenging classes and minimize confusion between visually similar ones is ongoing. This analysis will provide a valuable basis for the development team to strategize and make targeted improvements.

4.4. Test Set Strategy

Building upon the validation phase, where model performance exhibited stability and robustness across critical object classes, we identified the top-performing configurations. The selection was informed by peak performance metrics, consistency, and generalization capability, particularly in challenging categories. For the final test stage, we adopted a deployment strategy involving either the single most effective model or an ensemble of models exhibiting complementary behavior, aiming to mitigate class-specific weaknesses while preserving global accuracy.

Upon submission to the official contest platform, the selected model achieved a notable mean Average Precision at an IoU threshold of 0.50 (mAP@50) of 75.19% on the hidden test set. This result strongly supports the methodological choices in data preprocessing, architecture tuning, and training protocol. It underscores the model's practical efficacy and potential for real-world deployment in high-resolution remote sensing object detection tasks.

5. CONCLUSION

We presented an object detection approach for aerial cityscapes using YOLO variants and a targeted data augmentation pipeline. This strategy, focusing on class balance and instance enrichment, is designed to improve detection accuracy for the ICIP 2025 Challenge.

The strategy yielded auspicious results, achieving mAP@50 equals 75.19% (on the test data set) and securing rank 2³ in the ICIP 2025 competition. In the future, we plan to explore integrating generative AI models to refine and smooth edge cases further, enhancing the naturalness of the training data.

³<https://cadot.onrender.com/results>

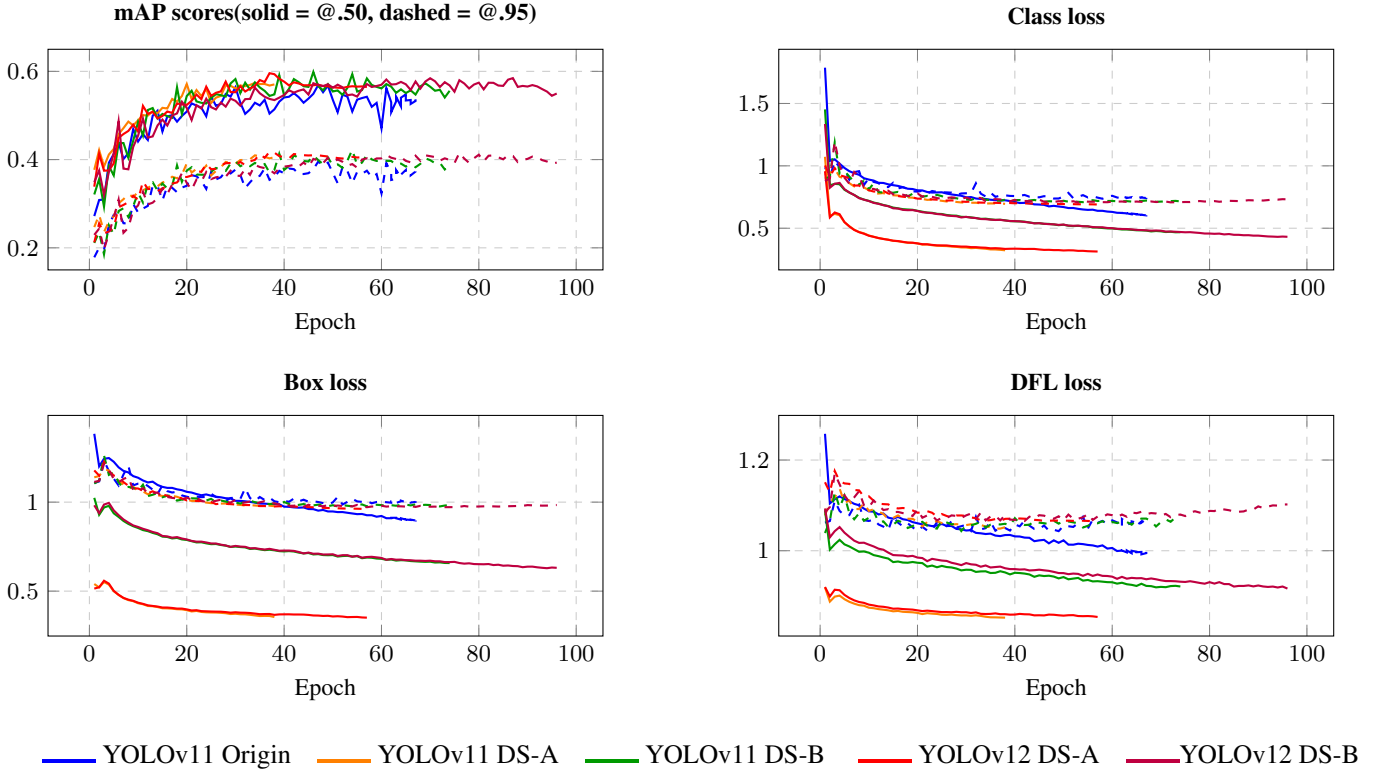


Fig. 4: Training dynamics across models. Solid = training; dashed = validation. Colors denote variants as shown in the legend above.

6. REFERENCES

- [1] Rahima Khanam and Muhammad Hussain, “Yolov11: An overview of the key architectural enhancements,” *arXiv preprint arXiv:2410.17725*, 2024.
- [2] Yunjie Tian, Qixiang Ye, and David Doermann, “Yolov12: Attention-centric real-time object detectors,” *arXiv preprint arXiv:2502.12524*, 2025.
- [3] Gui-Song Xia, Xiang Bai, Jian Ding, Zhen Zhu, Serge Belongie, Jiebo Luo, Mihai Datcu, Marcello Pelillo, and Liangpei Zhang, “Dota: A large-scale dataset for object detection in aerial images,” in *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2018, pp. 3974–3983.
- [4] Dawei Du, Longyin Wen, Pengfei Zhu, Heng Fan, Qinghua Hu, Haibin Ling, Mubarak Shah, Junwen Pan, Ali Al-Shuwaili, Amr Mohamed, Imene Bakour, Bin Dong, Binyu Zhang, Bouchali Nesma, Chenfeng Xu, Chenzhen Duan, Ciro Castiello, Corrado Mencar, Dingkan Liang, and Zhiyuan Zhao, *VisDrone-CC2020: The Vision Meets Drone Crowd Counting Challenge Results*, pp. 675–691, 08 2020.
- [5] Jian Ding, Nan Xue, Yang Long, Gui-Song Xia, and Qikai Lu, “Learning roi transformer for detecting oriented objects in aerial images,” *CoRR*, vol. abs/1812.00155, 2018.
- [6] Jiaming Han, Jian Ding, Nan Xue, and Gui-Song Xia, “Redet: A rotation-equivariant detector for aerial object detection,” *CoRR*, vol. abs/2103.07733, 2021.
- [7] Xingkui Zhu, Shuchang Lyu, Xu Wang, and Qi Zhao, “Tph-yolov5: Improved yolov5 based on transformer prediction head for object detection on drone-captured scenarios,” *CoRR*, vol. abs/2108.11539, 2021.
- [8] Zhiwei Wei and Chenzhen Duan, “Amrnet: Chips augmentation in aerial images object detection,” *CoRR*, vol. abs/2009.07168, 2020.
- [9] Ziyang Tang, Xiang Liu, Guangyu Shen, and Baijian Yang, “Penet: Object detection using points estimation in aerial images,” 01 2020.
- [10] Alexey Bochkovskiy, Chien-Yao Wang, and Hong-Yuan Mark Liao, “Yolov4: Optimal speed and accuracy of object detection,” *CoRR*, vol. abs/2004.10934, 2020.
- [11] Cuong Ly, Grayson Jorgenson, Dan Rosa de Jesus, Henry Kvinge, Adam Attarian, and Yijing Watkins, “Colmix – a simple data augmentation framework to improve object detector performance and robustness in aerial images,” *arXiv preprint arXiv:2305.13509*, 2023.
- [12] Martin Messmer, Benjamin Kiefer, and Andreas Zell, “Gaining scale invariance in UAV bird’s eye view object detection by adaptive resizing,” *CoRR*, vol. abs/2101.12694, 2021.
- [13] Marcelo Bertalmio, Guillermo Sapiro, Vincent Caselles, and Coloma Ballester, “Image inpainting,” in *Proceedings of the 27th Annual Conference on Computer Graphics and Interactive Techniques*, USA, 2000, SIGGRAPH ’00, p. 417–424, ACM Press/Addison-Wesley Publishing Co.